3-D Deformation Fields Construction Based on Spatial and Temporal Continuity for GNSS-Based InBSAR: Verified and Enhanced by GB-SAR

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***Abstract*— The 3-D deformation monitoring can be achieved by associating different monitoring results from different navigation satellites in Global Navigation Satellite System (GNSS)-based interferometric bistatic synthetic aperture radar (InBSAR) system. However, the whole scene monitoring accuracy is limited by the low signal-to-noise ratio (SNR) and the differences among synthetic aperture radar (SAR) images from different GNSS satellites. In this article, a modified 3-D deformation fields construction algorithm based on the spatial and temporal continuity is proposed for GNSS-based InBSAR system. First, based on the spatial continuity, the persistent scatterers (PSs) that can be observed by more satellites are used to filter the initial scene deformation monitoring results. Second, the model is established according to the temporal continuity of deformation to refine the optimal regularization parameters. Third, with the intersystem fusion concept, 1-D deformation with sub-millimeter accuracy of ground-based SAR (GB-SAR) is introduced to further improve the 3-D deformation monitoring accuracy of GNSS-based InBSAR. Finally, the experimental data verify the proposed algorithm for the GNSS-based InBSAR system, achieving millimeter-level deformation monitoring accuracy. The proposed algorithm enables the system have great potential in applications such as geological disaster warning and building health monitoring.**

***Index Terms*— 3-D deformation fields construction, Global Navigation Satellite System (GNSS)-based interferometric bistatic synthetic aperture radar (InBSAR), intersystem constraint, regularization parameter selection.**

I. Introduction

THE acquisition of high-accuracy 3-D deformation fields in scenes, such as potential hazard surfaces, highway slopes, and high-speed railway bridges is an important technical way to achieve geological disaster warning and building health monitoring. An important technical way to obtain 3-D millimeter-scale surface deformation data is the Global Navigation Satellite System (GNSS)-based interferometric bistatic synthetic aperture radar (GNSS-based InBSAR) technology [1], [2], [3]. This technique employs orbiting navigation satellites as external radiation sources, while a stationary receiver is positioned on the ground or installed on a vehicle or an aircraft [4], [5], [6], [7] to form a bistatic configuration. Then, the time-series synthetic aperture radar (SAR) images are generated for interferometric processing to measure the surface microdeformation.

Although GNSS satellites transmit coarse acquisition code rather than the chirp signal, there has been much research based on modulation signal focus and imaging [8], [9]. SAR images can also be obtained by using such signals. By utilizing the navigational signal, the GNSS-based InBSAR system has a series of advantages. First, this system has a higher temporal monitoring frequency. The repeat-pass period of the proposed system with the Beidou satellites in inclined geosynchronous orbit (IGSO) is 1 day, which is much shorter than that of the traditional interferometric SAR (InSAR) systems (for instance, the repeat-pass periods of TerraSAR-X, Sentinel-1, COSMO-SkyMed (single satellite), and Radarsat are 11 days, 6 days, 16 days, and 24 days, respectively [10], [11], [12], [13]). Second, for the Beidou navigation system, there are already 37 satellites [ten satellites in IGSO and 27 satellites in medium Earth orbit (MEO)] in orbit. Multiple GNSS transmitters ensure observation of the scene from multiple angles, so the 1-D deformation along different line of sight (LOS) of satellites can be obtained simultaneously. Then, the 3-D deformation can be calculated by associating different monitoring results from different navigation satellites [14]. Benefiting from these advantages, the GNSS-based InBSAR can be widely applied to monitor deformation.

However, there are some challenges in obtaining high-accuracy 3-D deformation fields due to the following reasons. First, the navigation satellites are designed for positioning, so only the received power of the direct signal is considered. The received power of the navigation signal by the 0-dB antenna is −165 dbW. After scene reflection and secondary propagation, the echo has large attenuation, resulting in a low signal-to-noise ratio (SNR) of the SAR images. This will further lead to poor interferometric phase accuracy and poor error compensation accuracy of the PSs. Subsequently,

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the 1-D deformation monitoring accuracy of each satellite will be deteriorated.

Second, by associating the 1-D monitoring results from different satellites [15], the 3-D deformation monitoring results can be obtained through the least squares method [1]. However, based on the position dilution of precision (PDOP) [14], high-3-D deformation monitoring results require the dispersed satellite distribution, i.e., larger differences of the incident angle. With the change in the incident angle, the radar cross section (RCS) of the scene will also change significantly. This will result in different distributions of the PSs in PS-InSAR technique. Particularly, a strong target can be monitored by one bistatic configuration; however, it has a low RCS under another bistatic configuration, which makes it cannot be monitored under this configuration. Therefore, different PSs are monitored by different satellite combinations, which means that the multistatic configurations are different, and further leads to different 3-D deformation monitoring accuracies. Therefore, the monitoring accuracy of the whole scene is uneven. What is more, it is inevitable that the elevation and azimuth angles [15] of the observing satellite of some PSs are particularly concentrated, making the observation matrix appear multicollinearity. Furthermore, it will result in the least squares estimation results inaccurate, i.e., the 3-D deformation monitoring results will be ill-posed, singular, and unstable. All of the mentioned challenges make it difficult to obtain high-accuracy 3-D surface deformation fields.

Current methods for measuring 3-D deformation include space-borne InSAR, ground-based SAR (GB-SAR), and fusion of them. As regards space-borne InSAR, the most straightforward strategy is integrating multitrack InSAR observations that are acquired from different imaging geometries and InSAR satellites [16], [17], [18]. In [17], 3-D deformation of the scene is obtained through multiple interferograms with four image geometries. In [18], the multisource SAR images from environmental satellite (ENVISAT) advanced SAR (ASAR) and advanced land observing satellite (ALOS) PalSAR are exploited to solve the 3-D deformation fields of the mining surface. As regards GB-SAR, the multistatic system based on three GB-SAR radars is proposed to realize 3-D deformation measurements. The feasibility of the system is verified through deformation measurements of a displaceable corner reflector [19] and an open-pit mine [20]. The multistatic image registration has been solved [21]. However, these methods have fewer input data dimensions, i.e., fewer transmitters, such as three GB-SAR radars and SAR satellites with ascending and descending geometries, which are relatively few. While the GNSS-based InBSAR system, with abundant in-orbit satellites, the input data have more dimensions, so the processing process is more complex. In addition, the low SNR will also decrease the accuracy of these monitoring algorithms.

Another strategy is combining InSAR measurements with other ancillary geodetic data, such as GNSS point positioning [22], [23]. The sparse GNSS points are interpolated into the same spatial grids of InSAR, and then, 3-D deformation fields are obtained by building a fusion model. In [22], the Markov random field based on regularization and simulated annealing optimization are used to merge the InSAR and interpolated GPS data to obtain high-resolution 3-D deformation. In [23], an iterative almost unbiased estimation algorithm is proposed to obtain the variance components of GNSS and InSAR. Moreover, another strategy imposes a prior information as the constraint on the deformation process, such as the surface-parallel flow assumption, i.e., movement along the steepest downslope [24]. However, the GNSS-based InBSAR system only needs a single receiver to achieve deformation measurement of surface scenes without requiring additional ancillary geodetic data, so these algorithms cannot be adopted. As regards intersystem fusion, a fusion model of 1-D deformation results monitored by GB-SAR and space-borne SAR model, respectively, is established in [25], and weighted least squares are used to solve the optimal 3-D deformation. However, the integration of GNSS-based InBSAR and GB-SAR systems combines 3-D and 1-D deformation. Therefore, the algorithm above is not suitable for the GNSS-based InBSAR system.

In conclusion, the low SNR and the changes of scene RCS make it difficult to obtain high-accuracy 3-D surface deformation fields in the GNSS-based InBSAR system. What is more, GNSS-based InBSAR has more input data dimensions and does not need additional ancillary geodetic data. To achieve high-accuracy 3-D deformation monitoring of the scene, the advantage of having a large number of satellites in orbit needs to be utilized. Therefore, the algorithms mentioned above proposed for space-borne SAR and GB-SAR are not suitable for GNSS-based InBSAR systems. In addition, they cannot obtain the 3-D deformation fields of the whole surface scene. While the GNSS-based InBSAR has great research potential in the field of surface deformation monitoring.

GNSS-based InBSAR technology has been gradually developed in recent years. There are abundant studies on imaging, resolution analysis, and so on [26], [27], [28], [29], [30], but less on deformation retrieval [1], [3]. The concept of GNSS-BiSAR system is proposed in [26]. In [27], [28], and [29], the range and azimuth resolutions and multistatic point spread function of bistatic SAR are analyzed. In [30], a method is proposed to obtain high-coherence image pairs for GNSS-based InSAR systems. The concept of GNSS-based InBSAR is introduced in [31] to achieve avalanche monitoring. In [3], the strong points are constructed artificially to simulate the surface deformation and achieve 1-D deformation monitoring of the targets with monitoring accuracy at the centimeter level. A fused image algorithm and interferometric phase error model of PSs are obtained in [32]. In [1], based on the transponder, 3-D deformation monitoring of a single point is achieved with monitoring accuracy at millimeter level. Later the experimental design [14], phase error compensation [33], multiangle images of different satellite association algorithms [15], [34], fake PSs removal algorithm [35], and the modified Kriging interpolator [36] are proposed to achieve higher monitoring efficiency and improve monitoring accuracy for the GNSS-based InBSAR system. Current research is primarily focused on 1-D and 3-D deformation monitoring of single points, yet there remains less attention on high-accuracy 3-D deformation monitoring of the whole scene.

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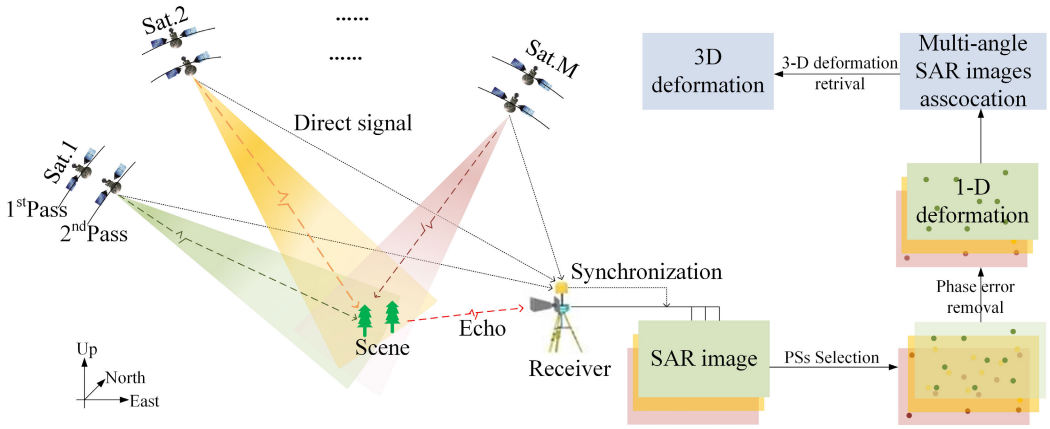


Fig. 1. Configuration and signal processing flow of GNSS-based InBSAR system. The transmitters are the navigation satellites in orbit, and the receiver is fixed on the ground. The isotropic antenna receives the direct signal to achieve synchronization, and the directional antenna receives the echo reflected from the scene. The east–north–up coordinate is used.

In this article, a 3-D deformation field construction algorithm is proposed for the GNSS-based InBSAR system. To address the problem of uneven deformation monitoring accuracy in the scene, a deformation field filtering method based on spatial and temporal continuity is proposed, which takes advantage of the abundance of satellites in the GNSS-based InBSAR system. In this way, it achieves high-accuracy 3-D deformation field construction of the scene without losing any observation information from any satellite. Furthermore, GB-SAR is introduced as the constraint to obtain higher accuracy 3-D deformation monitoring results in the GNSS-based InBSAR system. This cross-system approach provides more possibilities for the application of GNSS-based InBSAR. The proposed algorithm solves the problem that the whole scene monitoring accuracy is limited by the low SNR and the differences in SAR images from different satellites in GNSS-based InBSAR. The remainder of this article is organized as follows. In Section II, the 3-D deformation retrieval model of GNSS-based InBSAR is established. Section III gives a detailed explanation of the proposed algorithm. In Section IV, the feasibility of the proposed algorithm is verified by the raw GNSS-based InBSAR data, and the experimental results are shown step by step. Finally, Section V concludes this article.

II. Signal Model

In this section, the 3-D deformation retrieval model is proposed for GNSS-based InBSAR, and the faced challenges are analyzed carefully.

A. General Model of 3-D Deformation Retrieval

Different from traditional InSAR systems, in GNSS-based InBSAR systems, the transmitters are the navigation satellites in orbit, and the receiver is fixed on the ground, forming a bistatic configuration. The receiver contains two independent channels: the isotropic antenna and the directional antenna. The isotropic antenna receives the direct signal to achieve synchronization, and the directional antenna receives the echo reflected from the scene. The east–north–up coordinate is used and the receiver is located at the origin of the coordinate system. The key steps in GNSS-based InBSAR system to obtain 3-D deformation include SAR images formation, PS selection, phase error removal, multiangle SAR images association, and 3-D deformation retrieval [1], [14], [33], as shown in Fig. 1. The specific process is as follows.

In the GNSS-based InBSAR system, the PS-InSAR technique is used to achieve deformation retrieval. For a certain PS q observed by any satellite, on the day n, the phase of the SAR image can be expressed as follows:

$$\begin{aligned}\varphi^{(n)}(q) &= 2\pi \times \frac{\left|\mathbf{PS}^{(n)}-\mathbf{PQ}^{(n)}(q)\right|+\left|\mathbf{PE}-\mathbf{PQ}^{(n)}(q)\right|-\left|\mathbf{PS}^{(n)}-\mathbf{PE}\right|}{\lambda_L} \\ &\quad + \varphi_{\text{noise}}^{(n)}(q) \end{aligned} \tag{1}$$

where λ_L is the wavelength of L-band, **PS** is the central position of the satellite trajectory, **PQ** is the position of the target, **PE** is the position of the receiver, they are all 3×1 vectors, and $\varphi_{\text{noise}}^{(n)}(q)$ is the phase noise.

Let $n-1$ day be the reference time, then the interferometric phase of PS q between n day and $n-1$ day can be denoted as follows:

$$\begin{aligned}\Delta\varphi^{(n)}(q) &= \varphi^{(n)}(q) - \varphi^{(n-1)}(q) \\ &= \frac{2\pi}{\lambda_L} \times \left\{ \begin{array}{l} \left|\mathbf{PS}^{(n)} - \mathbf{PQ}^{(n)}(q)\right| + \left|\mathbf{PE} - \mathbf{PQ}^{(n)}(q)\right| \\ -\left|\mathbf{PS}^{(n)} - \mathbf{PE}\right| + \left|\mathbf{PS}^{(n-1)} - \mathbf{PQ}^{(n-1)}(q)\right| \\ +\left|\mathbf{PE} - \mathbf{PQ}^{(n-1)}(q)\right| - \left|\mathbf{PS}^{(n-1)} - \mathbf{PE}\right| \end{array} \right\} \\ &\quad + \Delta\varphi_{\text{noise}}^{(n)}(q) \\ &= \Delta\varphi_{\text{defo}}^{(n)}(q) + \Delta\varphi_{\text{topo}}^{(n)}(q) + \Delta\varphi_{\text{noise}}^{(n)} \end{aligned} \tag{2}$$

where $\Delta\varphi_{\text{defo}}^{(n)}(q)$ is the interferometric phase caused by the deformation of PS, $\Delta\varphi_{\text{topo}}^{(n)}(q)$ is the topographic phase, and $\Delta\varphi_{\text{noise}}^{(n)}(q)$ is the phase noise, such as thermal noise, atmospheric phase error, and channel phase error.

Then, the topographic phase is compensated [1]. Because the position of q is unknown, q' which is the focus position

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of q is used to calculate the topographic phase

$$\Delta\varphi_{\text{topo}}^{(n)}(q) \approx \frac{2\pi}{\lambda_L}\left[\mathbf{PS}^{(n)}-\mathbf{PS}^{(n-1)}\right]^{\mathrm{T}} \times\left(\frac{\mathbf{PS}^{(n-1)}-\mathbf{PQ}^{(n-1)}(q')}{\left|\mathbf{PS}^{(n-1)}-\mathbf{PQ}^{(n-1)}(q')\right|}-\frac{\mathbf{PS}^{(n-1)}-\mathbf{P}_{\mathbf{E}}}{\left|\mathbf{PS}^{(n-1)}-\mathbf{P}_{\mathbf{E}}\right|}\right). \quad (3)$$

After the topographic phase is removed, the residual interferometric phase can be described as follows:

$$\begin{aligned}\Delta\varphi'^{(n)}(q) &= \Delta\varphi_{\text{noise}}'^{(n)}(q)-\frac{2\pi}{\lambda_L}\left[\Delta\mathbf{PQ}^{(n)}(q)\right]^{\mathrm{T}} \times\left(\frac{\mathbf{PS}^{(n)}-\mathbf{PQ}^{(n-1)}(q)}{\left|\mathbf{PS}^{(n)}-\mathbf{PQ}^{(n-1)}(q)\right|}+\frac{\mathbf{PE}-\mathbf{PQ}^{(n-1)}(q)}{\left|\mathbf{PE}-\mathbf{PQ}^{(n-1)}(q)\right|}\right) \\ &= \Delta\varphi_{\text{noise}}'^{(n)}(q)+\Delta\varphi_{\text{defo}}^{(n)}(q) \end{aligned} \quad (4)$$

where $\Delta\varphi_{\text{noise}}'^{(n)}(q)$ is the phase noise which contains elevation phase error, channel phase error, atmospheric phase error, and thermal noise.

Then, the phase noise is compensated [30], [33], and the residual interferometric phase is

$$\begin{aligned}\Delta\varphi''^{(n)}(q) &= \Delta\varphi_{\text{defo}}^{(n)}(q)\approx-\frac{2\pi}{\lambda_L}\left[\Delta\mathbf{PQ}^{(n)}(q)\right]^{\mathrm{T}} \times\left(\frac{\mathbf{PS}^{(n)}-\mathbf{PQ}^{(n-1)}(q)}{\left|\mathbf{PS}^{(n)}-\mathbf{PQ}^{(n-1)}(q)\right|}+\frac{\mathbf{PE}-\mathbf{PQ}^{(n-1)}(q)}{\left|\mathbf{PE}-\mathbf{PQ}^{(n-1)}(q)\right|}\right) \end{aligned} \quad (5)$$

where $\Delta\mathbf{PQ}^{(n)}(q)=\mathbf{PQ}^{(n)}(q)-\mathbf{PQ}^{(n-1)}(q)$.

Afterward, multiangle SAR images association is performed to obtain PSs observed by multiple satellites [15]. Then, the 3-D deformation model can be established as follows:

$$\mathbf{\Phi}_{M\times1}^{n}(q)=\mathbf{H}_{M\times3}^{n}(q)\times\mathbf{D}_{3\times1}^{n}(q)+\mathbf{n}_{M\times1} \quad (6)$$

where $\mathbf{\Phi}$, $\mathbf{H}$, and $\mathbf{D}$ are shown in (7), as shown at the bottom of the page. M, N, and Q are the number of observing satellites, days, and PSs.

Subsequently, the least squares method is used to estimate 3-D deformation, and the objective function is

$$J\left(\mathbf{D}^{n}\right)=\left\|\mathbf{\Phi}^{n}-\mathbf{H}^{n}\cdot\mathbf{D}^{n}\right\|^{2}. \quad (8)$$

Therefore, the 3-D deformation could be retrieved by using the Moore–Penrose generalized inverse matrix method

$$\widehat{\mathbf{D}}_{\text{ls}}^{n}(q)=\left((\mathbf{H}^{n})^{\mathrm{T}}\mathbf{H}^{n}\right)^{-1}(\mathbf{H}^{n})^{\mathrm{T}}\mathbf{\Phi}^{n} \quad (9)$$

where $((\mathbf{H}^{n})^{\mathrm{T}}\mathbf{H}^{n})^{-1}(\mathbf{H}^{n})^{\mathrm{T}}$ is the 3-D mapping operator.

Ultimately, the deformation monitoring accuracy can be calculated as follows:

$$\mathbf{Acc}=\sqrt{\frac{1}{Q}\sum_{q=1}^{Q}\left(\widehat{\mathbf{D}}_{\text{ls}}^{n}(q)-\mathbf{D}_{\text{nomi}}^{n}(q)\right)^{2}},\quad n=2,\ldots,N \quad (10)$$

where **Acc** is the accuracy of east, north, and up directions. $\mathbf{D}_{\text{nomi}}$ is the nominal deformation of PSs.

B. Problem Analysis

However, there are some challenges in obtaining the 3-D deformation fields. The main problems include the low SNR, ill-posed problem, and uneven number of associated satellites, which are further explained as follows.

1) Compared with the traditional InSAR systems, a navigation signal is adopted in the GNSS-based InBSAR system. The received power of the navigation signal by the 0-dB antenna is −165 dbW. After scene reflection and secondary propagation, the echo has a large attenuation, resulting in the low SNR of the SAR images. Therefore, the interferometric phase accuracy and error compensation accuracy of the PSs are poor, which deteriorates the 1-D deformation monitoring accuracy of each satellite. Fig. 2 shows a single satellite's deformation observation results of ten PSs under the condition of no displacement. It can be seen that due to the influence of low SNR, the deformation results measured by a single satellite are low.
2) By associating the 1-D monitoring results from different satellites [15], the 3-D deformation monitoring results can be obtained through the least squares method [1]. However, based on the PDOP [14], high 3-D deformation monitoring results require a dispersed satellite distribution, i.e., a larger difference of the incident angle of satellites. With the change of the incident angle,

$$\mathbf{\Phi}_{M\times1}^{n}(q)=\begin{bmatrix}\Delta\varphi_{1}^{(n)}(q)\\ \Delta\varphi_{2}^{(n)}(q)\\ \vdots\\ \Delta\varphi_{M}^{(n)}(q)\end{bmatrix},\quad \mathbf{H}_{M\times3}^{n}(q)=-\frac{2\pi}{\lambda_L}\begin{bmatrix}\frac{\left[\mathbf{PS}_{1}^{(n)}-\mathbf{PQ}^{(n)}(q)\right]^{\mathrm{T}}}{\left|\mathbf{PS}_{1}^{(n)}-\mathbf{PQ}^{(n)}(q)\right|}+\frac{\left[\mathbf{PE}-\mathbf{PQ}^{(n)}(q)\right]^{\mathrm{T}}}{\left|\mathbf{PE}-\mathbf{PQ}^{(n)}(q)\right|}\\ \vdots\\ \frac{\left[\mathbf{PS}_{M}^{(n)}-\mathbf{PQ}^{(n)}(q)\right]^{\mathrm{T}}}{\left|\mathbf{PS}_{M}^{(n)}-\mathbf{PQ}^{(n)}(q)\right|}+\frac{\left[\mathbf{PE}-\mathbf{PQ}^{(n)}(q)\right]^{\mathrm{T}}}{\left|\mathbf{PE}-\mathbf{PQ}^{(n)}(q)\right|}\end{bmatrix}$$

$$\mathbf{D}_{3\times1}^{n}(q)=\begin{bmatrix}\mathbf{D}_{x}^{(n)}(q)\\ \mathbf{D}_{y}^{(n)}(q)\\ \mathbf{D}_{z}^{(n)}(q)\end{bmatrix}=\Delta\mathbf{PQ}^{(n)}(q),\quad n=2,\ldots,N;\ q=1,\ldots,Q;\ M\geq3 \quad (7)$$

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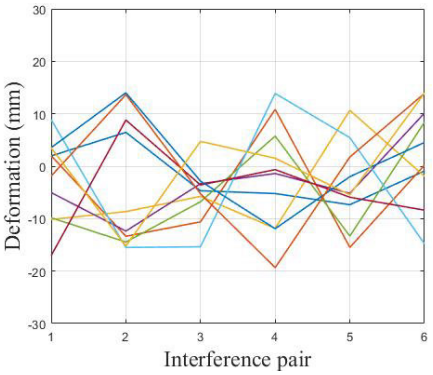


Fig. 2. Deformation monitoring results of ten PSs measured by a single satellite.

TABLE I
THREE-DIMENSIONAL DEFORMATION MONITORING ACCURACY AND NUMBER OF PSS MONITORED BY SEVEN BEIDOU SATELLITES IN IGSO

Number of observing satellites	Accuracy (mm)			Number of PSs
	East	North	Up	
3	83.67	81.62	102.31	98
4	45.53	23.56	39.62	82
5	22.13	12.81	14.22	68
6	6.83	5.72	6.17	40
7	5.41	3.87	5.81	13
All of the PSs	48.72	35.03	43.63	301

the RCS of the scene will also change significantly. This will result in a different distribution of the PSs in PS-InSAR technique. Moreover, different PSs are monitored by different satellite combinations, i.e., M in (7), and the multistatic configurations are different. These result in the surface deformation monitoring accuracy uneven on the monitoring scene. Table I shows the 3-D deformation accuracy corresponding to different numbers of satellites. It is obvious that the deformation accuracy of PSs observed from 6 to 7 satellites is within 10 mm, which is much better than that of PSs observed from 3 to 5 satellites. What is more, the distribution of 3-D deformation monitoring accuracy of the whole scene is uneven. This leads to the average accuracy of the entire scene being only about 40 mm.

3) In addition, the accuracy of the least-squares solution is related to the dispersion degree of the input data. When the amount of input data is small and the data distribution is relatively concentrated, the ill-posed problem will be introduced and the least-squares solution may be singular. Because the satellites observing the PSs are random, if the elevation and azimuth angles of observing satellites are relatively similar, $\mathbf{H}$ in (9) will appear multicollinearity, and the regression equation will be an ill-posed problem. The deformation retrieval result $\widehat{\mathbf{D}}_{\text{ls}}$ will be ill-conditioned and singular.

Therefore, in order to mitigate the above problems, an algorithm to obtain high-accuracy 3-D deformation fields for the GNSS-based InBSAR system is required.

III. MODIFIED 3-D DEFORMATION RETRIEVAL ALGORITHM

In this section, a modified 3-D deformation retrieval algorithm is proposed. First, based on the spatial continuity, the PSs that can be observed by more satellites are used to filter the initial scene deformation monitoring results. Then, the optimal regularization parameter is refined according to the temporal continuity of deformation. So far, high-accuracy deformation fields can be achieved based on the GNSS-based InBSAR system by itself. On this basis, with the intersystem fusion concept, 1-D deformation with sub-millimeter accuracy of GB-SAR is introduced to further improve the 3-D deformation monitoring accuracy of GNSS-based InBSAR. The whole flowchart is shown in Fig. 3.

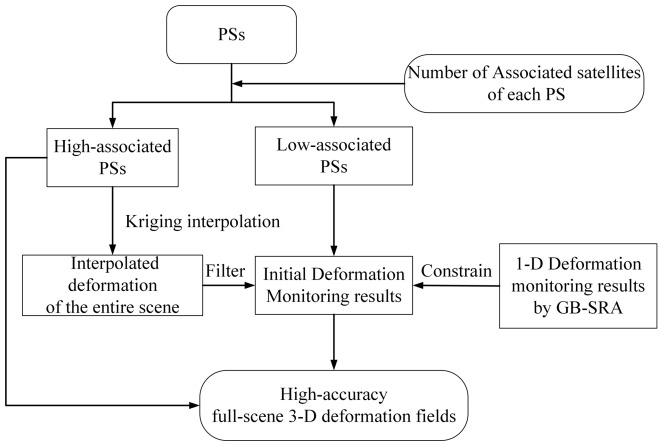


Fig. 3. Flowchart of the proposed high-accuracy 3-D deformation fields construction algorithm.

A. Optimization Model

As mentioned above, the deformation accuracy of each PS is related to the number of satellites observing it. Specifically, it can be seen from Table I, when the number of observing satellites reach up to 6, the 3-D deformation monitoring accuracy is below 10 mm. In contrast, when the number of observing satellites does not reach up 6, the accuracy deteriorates a lot. It is hoped that all of the PSs can reach monitoring accuracy within 10 mm. To achieve this goal, we establish the optimization model.

First, let the set of all PSs be $\bar{G} = [G_3, G_4, \ldots, G_M]$, and divide them according to the number of associated satellites

$$\begin{aligned} \bar{G}_b &= [G_3, \ldots, G_S] \\ \bar{G}_f &= [G_{S+1}, \ldots, G_M] \\ \bar{G} &= \bar{G}_b \cup \bar{G}_f \end{aligned} \tag{11}$$

where $\bar{G}_b$ and $\bar{G}_f$ denote the sets of PSs with lower and higher number of associated satellites, respectively. $G_3, \ldots, G_M$ represent the PSs with $3, \ldots, M$ associated satellites. S represents the threshold separating low and high associated satellites.

Due to the spatial continuity of deformation, the high-associated point set can be used to interpolate the deformation of the whole scene. During the interpolation process, the influence of points closer to the target is increased, and vice versa. Considering the Kriging interpolation is a typical inverse distance weighted interpolation method [37], it is adopted here, as shown in Fig. 4, where red, green, and yellow points represent the target, high-associated, and low-associated PSs, respectively. The interpolated deformation at the target

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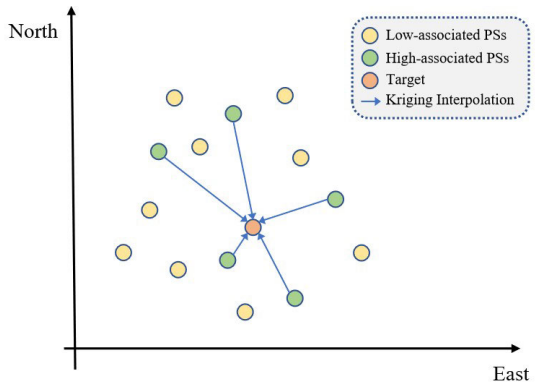


Fig. 4. Schematic of kriging interpolation. The interpolated deformation at the target point can be obtained through kriging interpolation of green points.

point can be obtained through kriging interpolation of green points. Taking a low-associated PS (x_0, y_0) as an instance, the corresponding interpolated deformation in the east, north, and up directions can be obtained

$$\begin{aligned}\bar{\mathbf{D}}(x_0, y_0) &= \text{kriging}\left(\hat{\mathbf{D}}_{ls}\left(\bar{G}_f\right)\right) \\ &= \sum_{i=1}^{Q_2} \lambda_i(x_0, y_0)\hat{\mathbf{D}}_{ls}(s_i) \\ &= \left[\bar{D}_{\text{east}}(x_0, y_0), \bar{D}_{\text{north}}(x_0, y_0), \bar{D}_{\text{up}}(x_0, y_0)\right] \\ &\qquad (x_0, y_0) \in \bar{G}_b, \quad s_i \in \bar{G}_f \end{aligned} \tag{12}$$

where Q_2 is the number of high-associated PSs and λ_i is the coefficient which is related to the distance between the target point and the high-associated PSs.

Furthermore, the interpolated deformation fields can be used to filter the deformation monitoring results of low-associated PSs in $\bar{G}_b$. Compared with (8), the improved optimization objective function is established

$$J(\mathbf{D}_{\text{cls}}, k_1) = \|\mathbf{\Phi} - \mathbf{H}\cdot\mathbf{D}_{\text{cls}}\|_2^2 + k_1\frac{2\pi}{\lambda_L}\left(\left\|\mathbf{D}_{\text{cls}} - \bar{\mathbf{D}}\right\|_2^2\right) \tag{13}$$

where $\bar{\mathbf{D}}$ is the kriging interpolation results in (12). k_1 is the regularization parameter to control the balance between the data fitting and the regularization term in the objective function.

The optimal 3-D deformation estimation result $\hat{\mathbf{D}}_{\text{cls}}$ is the solution that minimizes the objective function (13), which can be denoted as follows:

$$\begin{aligned}\hat{\mathbf{D}}_{\text{cls}} &= \left(\mathbf{H}^{\text{T}}\mathbf{H} + \frac{2\pi}{\lambda_L}k_1\mathbf{I}\right)^{-1}\left(\mathbf{H}^{\text{T}}\mathbf{\Phi} + \frac{2\pi}{\lambda_L}k_1\bar{\mathbf{D}}\right) \\ &= \left(\mathbf{H}^{\text{T}}\mathbf{H} + \frac{2\pi}{\lambda_L}k_1\mathbf{I}\right)^{-1}\mathbf{H}^{\text{T}}\left(\mathbf{\Phi} - \mathbf{H}\bar{D}\right) + \bar{\mathbf{D}}. \end{aligned} \tag{14}$$

Finally, the 3-D deformation fields are represented as follows:

$$\hat{\mathbf{D}}_{\text{all}}(x, y) = \text{kriging}\left(\hat{\mathbf{D}}_{\text{cls}}\left(\bar{G}_b\right) \cup \hat{\mathbf{D}}_{ls}\left(\bar{G}_f\right)\right) \tag{15}$$

where (x, y) represents any point in the scene.

B. Regularization Parameter k_1 Selection

It is obvious that the regularization parameter k_1 is introduced in (13). In this section, the selection of k_1 is mainly discussed. Based on (13), different values of k_1 determine different degrees of filtering and lead to different deformation monitoring results. A smaller k_1 cannot completely solve the ill-posed problem, so the deformation monitoring accuracy of the PSs in $\bar{G}_b$ cannot be significantly improved. While a larger k_1 may introduce a larger monitoring error when a landslide is imminent, which means that a large-scale deformation is occurred. Therefore, it is important to choose an optimal regularization parameter k_1.

Particularly, the selection of k_1 is related to the multistatic configuration reflected in $\mathbf{H}$ and time-series deformation. Based on (14), the 3-D deformation is calculated from $\mathbf{H}$. Different PSs are observed by different satellites, and their positions relative to the receiver are also different, so they have different bistatic configurations. These factors make $\mathbf{H}$ and the corresponding col-linearity degree of $\mathbf{H}$ different for each PS. Therefore, different PSs have different optimal k_1. In addition, the current studies indicate that the deformation is continuous in time series [38], [39], [40]. Consequently, the deformation of one day can be estimated based on the deformation of the previous period and further obtain the optimal value of k_1.

But according to (7), $\mathbf{H}$ is a matrix determined by the multistatic configuration, and its dimension is related to the number of satellites observing the PSs. The deformation $\hat{\mathbf{D}}_{\text{all}}$ of PS is a vector with the dimension of 3×1, which has great randomness due to be affected by many factors such as noise. It is obvious that k_1, $\mathbf{H}$, and $\hat{\mathbf{D}}_{\text{all}}$ have different dimensions and influencing factors. Therefore, it is difficult to establish a theoretical model between them.

However, the matrix $\mathbf{H}$ and time-series deformation $\hat{\mathbf{D}}_{\text{all}}^{n-1}$ of each PS can be obtained. Therefore, instead of quantitative calculation, the numerical solution, i.e., simulation, can be used to obtain the selection results of k_1 of different PSs at different times. The specific process is as follows.

For a low-associated PS q, its position is $\mathbf{P}_q$ and the effective deformation monitoring point set in its neighboring area is defined as follows:

$$\mathbb{R}_{\text{cls}}(q) = \left\{B \mid \left\|\mathbf{P}_q, \mathbf{B}\right\|_2 < r_{\text{cls}}, B \in \bar{G}\right\} \tag{16}$$

where $\mathbb{R}_{\text{cls}}(q)$ is a point set, r_{cls} represents the radius, $\mathbf{B}$ represents the position of PS B, $\|\|_2$ is the two-norm which represents the distance between $\mathbf{B}$ and $\mathbf{P}_q$.

For each point in the point set $\mathbb{R}_{\text{cls}}$, their estimated deformation results $\hat{\mathbf{D}}_{\text{cls}}$ and interpolated deformation results $\bar{\mathbf{D}}$ of the previous $n-1$ days are known. Therefore, $\mathbf{D}_{\text{diff}} = \hat{\mathbf{D}}_{\text{cls}} - \bar{\mathbf{D}}$ of $n-1$ days can be calculated. Here, it is considered that the difference is continuous, so the $\hat{\mathbf{D}}_{\text{cls}} - \bar{\mathbf{D}}$ on nth day can be calculated by second-order fitting

$$\tilde{\mathbf{D}}_{\text{diff}}^{n}(p) = \text{polyval}\left(\underset{i=2}{\overset{n-1}{\text{polyfit}}}\left(\hat{\mathbf{D}}_{\text{all}}^{i}(p) - \bar{\mathbf{D}}^{i}(p)\right), n\right) \quad p \in \mathbb{R}_{\text{cls}}(q) \tag{17}$$

where polyval$(\cdot)$ and polyfit$(\cdot)$ represent the second-order fitting and valuation operations of polynomials. It should be

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noted that $-$, $\sim$, and $\wedge$ represent the interpolated, expected (simulated), and estimated values, respectively.

Then, $\tilde{\mathbf{D}}^n_{\text{diff}}$ of the target low-associated PS q can be calculated as follows:

$$\tilde{\mathbf{D}}^n_{\text{diff}}(q) = \frac{\sum_{i=1}^{P'} \mathbf{D}^n_{\text{diff}}(p_i)}{P'}, \quad p \in \mathbb{R}_{\text{cls}}(q) \tag{18}$$

where q is the target low-associated PS, p is the PS in $\mathbb{R}_{\text{cls}}(q)$, and P' is the number of $\mathbb{R}_{\text{cls}}(q)$.

Therefore, on the nth day, the expected deformation of PS q is

$$\tilde{\mathbf{D}}^n_{\text{cls}}(q) = \tilde{\mathbf{D}}^n_{\text{diff}}(q) + \bar{\mathbf{D}}^n(q). \tag{19}$$

Here, the expected deformation results are obtained based on temporal and interpolated deformation. It is derived from temporal predictions and does not incorporate the InSAR measurement results for the current day. Simultaneously, it can be considered that the 3-D deformation results of low-associated points in $\bar{G}_b$ measured by GNSS-based InBSAR should be close to the results obtained by temporal prediction. Based on (13), $\mathbf{H}^n$ and $\bar{\mathbf{D}}^n$ are known, and $\tilde{\mathbf{D}}^n_{\text{cls}}$ has been predicted. Therefore, if k_1 can be found that makes $\hat{\mathbf{D}}^n_{\text{cls}}$ closest to the expected result $\tilde{\mathbf{D}}^n_{\text{cls}}$, this k_1 is considered to be the most suitable.

Next, the value of k_1 is determined through numerical simulation. Taking $\tilde{\mathbf{D}}^n_{\text{cls}}(q)$ as the input, the simulated observed interferometric phase of each satellite can be expressed as follows:

$$\tilde{\boldsymbol{\Phi}}^n(q) = \mathbf{H}^n(q) \times \tilde{\mathbf{D}}^n_{\text{cls}}(q) + \boldsymbol{\xi} \tag{20}$$

where the observation matrix $\mathbf{H}^n(q)$ for q is known and $\boldsymbol{\xi}$ is the Gaussian noise which can be decided by the SNR of q. Up to now, all the parameters in (13) have been obtained, and the next step is to find the k_1 corresponding to the minimal (13).

Set the upper limit of parameter k_1 to $k_{1\max}$, let $\tilde{k}_1(q) \in [0, k_{1\max}]$. For each $\tilde{k}_1$, calculate the 3-D deformation estimation results by (14)

$$\overleftrightarrow{\mathbf{D}}^n_{\text{cls}}(q) = \left(\mathbf{H}^n(q)^{\mathrm{T}}\mathbf{H}^n(q) + \frac{2\pi}{\lambda_L}\tilde{k}_1(q)\mathbf{I}\right)^{-1} \times \left(\mathbf{H}^n(q)^{\mathrm{T}}\tilde{\boldsymbol{\Phi}}^n(q) + \frac{2\pi}{\lambda_L}\tilde{k}_1(q)\bar{\mathbf{D}}^n(q)\right). \tag{21}$$

Furthermore, the difference between $\overleftrightarrow{\mathbf{D}}^n_{\text{cls}}(q)$ and the expected output result $\tilde{\mathbf{D}}^n_{\text{cls}}(q)$ can be expressed as follows:

$$\text{err}^n(q) = \left\| \overleftrightarrow{\mathbf{D}}^n_{\text{cls}}(q) - \tilde{\mathbf{D}}^n_{\text{cls}}(q) \right\|_2. \tag{22}$$

When the output result is closest to the expected output result, as well as $\text{err}^n(q)$ is minimum, the corresponding regularization parameter is optimal

$$\hat{k}^n_1(q) = \underset{k_1}{\text{argmin}} \ \left(\text{err}^n(q)\right). \tag{23}$$

It can be seen from (20), because there is a random noise $\boldsymbol{\xi}$ in the simulation of $\tilde{\boldsymbol{\Phi}}^n(q)$, the Monte Carlo method is used to determine the final value of k_1. Regenerate the noise $\boldsymbol{\xi}$ randomly and repeat the above operation. Then, the mode of $\hat{k}^n_1(q)$ is chosen as the final $\hat{k}^n_{1\text{mode}}(q)$. Repeat the above operation for each PS and the regularization parameter k_1 of all low-associated PSs on the day n can be calculated. Finally, the deformation monitoring results on any day after corrected can be obtained

$$\begin{aligned}\widehat{\mathbf{D}}_{\text{cls}} &= \left(\mathbf{H}^{\mathrm{T}}\mathbf{H} + \frac{2\pi}{\lambda_L}\hat{k}_{1\text{mode}}\mathbf{I}\right)^{-1}\left(\mathbf{H}^{\mathrm{T}}\boldsymbol{\Phi} + \frac{2\pi}{\lambda_L}\hat{k}_{1\text{mode}}\bar{\mathbf{D}}\right) \\ &= \left(\mathbf{H}^{\mathrm{T}}\mathbf{H} + \frac{2\pi}{\lambda_L}\hat{k}_{1\text{mode}}\mathbf{I}\right)^{-1}\mathbf{H}^{\mathrm{T}}\left(\boldsymbol{\Phi} - \mathbf{H}\bar{\mathbf{D}}\right) + \bar{\mathbf{D}}. \end{aligned} \tag{24}$$

Through numerical simulation, we achieve to select the regularization parameter k_1 for PSs from different positions. In addition, benefiting from utilizing prior temporal information to choose k_1 not only confines the least squares results within a credible range but also ensures the monitoring accuracy of deformation.

C. Intersystem Constraint

GB-SAR can monitor high-accuracy deformation in the LOS. The fusion of the two systems can effectively improve the 3-D deformation monitoring accuracy of the GNSS-based InBSAR system. Therefore, the GBSAR high-accuracy radial deformation monitoring results are introduced as the constraint to further improve the 3-D deformation monitoring results of GNSS-based InBSAR.

The deformation observing results by GB-SAR can be expressed as follows:

$$\mathbf{D}_{\text{GB}} = |\mathbf{D}_{\text{GB}}|\mathbf{e}_{\text{PE}} = \frac{\lambda_{\text{Ku}}}{4\pi}\varphi_{\text{GB}} \cdot \mathbf{e}_{\text{PE}} \tag{25}$$

where $\mathbf{e}_{\text{PE}}$ is the unit vector from the PS to the receiver, λ_{Ku} is the wavelength of Ku-band and φ_{GB} is the interferometric phase observed by GB-SAR. Then, based on the coordinate system transformation and the projection principle of monostatic and bistatic SAR imaging, the SAR image registration between GNSS-based InBSAR and GB-SAR systems can be achieved [21].

Afterward, taking the 1-D high-accuracy deformation measurements of GB-SAR as the constraint, the optimization objective function (13) can be changed as follows:

$$\begin{aligned} J(\mathbf{D}_{\text{cls2}}, k_1, k_2) = \|\boldsymbol{\Phi} - \mathbf{H}\cdot\mathbf{D}_{\text{cls2}}\|_2^2 + k_1\frac{2\pi}{\lambda_L}\left(\left\|\mathbf{D}_{\text{cls2}} - \bar{\mathbf{D}}\right\|_2^2\right) \\ + k_2\frac{4\pi}{\lambda_{\text{Ku}}}\left(\|\mathbf{G}\cdot\mathbf{D}_{\text{cls2}} - |\mathbf{D}_{\text{GB}}|\|_2^2\right) \end{aligned} \tag{26}$$

where k_1 and k_2 are the regulation parameters. $\mathbf{G}_{1\times 3}$ is the transformation matrix, by using it, the 3-D GNSS-based InBSAR monitoring results can be projected to 1-D (along the LOS of GBSAR). The transformation matrix is equal to $\mathbf{e}_{\text{PE}}$.

Find the optimal 3-D deformation observation result $\hat{\mathbf{D}}_{\text{cls2}}$ that minimizes the objective function

$$\widehat{\mathbf{D}}_{\text{cls2}} = \arg\min_{\mathbf{D}_{\text{cls2}}} \{J(\mathbf{D}_{\text{cls2}}, k_1, k_2)\}. \tag{27}$$

Solving the equation, $\widehat{\mathbf{D}}_{\text{cls2}}$ can be expressed as follows:

$$\widehat{\mathbf{D}}_{\text{cls2}} = \left(\mathbf{H}^{\mathrm{T}}\mathbf{H} + \frac{2\pi}{\lambda_L}k_1\mathbf{I} + \frac{4\pi}{\lambda_{\text{Ku}}}k_2\mathbf{G}^{\mathrm{T}}\mathbf{G}\right)^{-1} \times \left(\mathbf{H}^{\mathrm{T}}\boldsymbol{\Phi} + \frac{2\pi}{\lambda_L}k_1\bar{\mathbf{D}} + \frac{4\pi}{\lambda_{\text{Ku}}}k_2|\mathbf{D}_{\text{GB}}|\mathbf{G}^{\mathrm{T}}\right). \tag{28}$$

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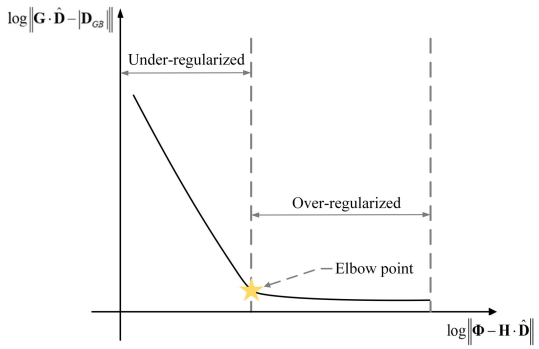


Fig. 5. L-curve diagram. Choose a set of regularization parameters $[0, k_{2\max}]$, solve the regression problem and get the estimated deformation $\hat{\mathbf{D}}$ versus each k_2. The L-curve is fit by many points ($\log\|\mathbf{\Phi} - \mathbf{H}\cdot\hat{\mathbf{D}}\|$, $\log\|\mathbf{G}\cdot\hat{\mathbf{D}} - |\mathbf{D}_{\mathrm{GB}}|\|$). k_2 corresponding to the elbow point of the L-curve is the optimal regularization parameter.

The selection method of k_1 has been proposed in Section III-B, so only k_2 needs to be determined here. The L-curve method is used to choose the regularization parameter k_2 [41], [42]. Let $k_2 \in [0, k_{2\max}]$ and calculate (28), the results of $\log\|\mathbf{\Phi} - \mathbf{H}\cdot\hat{\mathbf{D}}\|$ and $\log\|\mathbf{G}\cdot\hat{\mathbf{D}} - |\mathbf{D}_{\mathrm{GB}}|\|$ are shown in Fig. 5.

It can be found that $\log\|\mathbf{\Phi} - \mathbf{H}\cdot\hat{\mathbf{D}}\|$ and $\log\|\mathbf{G}\cdot\hat{\mathbf{D}} - |\mathbf{D}_{\mathrm{GB}}|\|$ are negatively related. The purpose of (27) is to find a k_2 that minimizes both of them at the same time. There is a clear elbow point that satisfies this requirement, as shown by the yellow dot in Fig. 5. This point can be identified by locating the point on the curve, where the curvature is maximum.

The curvature of the L-curve is

$$\kappa(k_2) = 2\frac{\hat{\rho}'\hat{\eta}'' - \hat{\rho}''\hat{\eta}'}{\left(\left(\hat{\rho}'\right)^2 + \left(\hat{\eta}'\right)^2\right)^{2/3}} \tag{29}$$

where $\hat{\eta} = \|\hat{\mathbf{D}}\|$, $\hat{\rho} = \|\mathbf{\Phi} - \mathbf{H}\cdot\hat{\mathbf{D}}\|$, and $'$ and $''$ mean the first and second derivatives.

Find the maximum value of (29), and the corresponding point is the corner of the L-curve

$$\hat{k}_2 = \arg\max_{k_2} \ (\kappa). \tag{30}$$

This approach helps us to find a suitable value of k_2 automatically. Therefore, the optimized 3-D deformation $\hat{\mathbf{D}}_{\mathrm{cls2}}$ is the solution that minimizes the objective function (26), which can be denoted as follows:

$$\begin{aligned}\hat{\mathbf{D}}_{\mathrm{cls2}} = &\left(\mathbf{H}^{\mathrm{T}}\mathbf{H} + \frac{2\pi}{\lambda_L}\hat{k}^n_{\mathrm{mode}}\mathbf{I} + \frac{4\pi}{\lambda_{\mathrm{Ku}}}\hat{k}_2\mathbf{G}^{\mathrm{T}}\mathbf{G}\right)^{-1} \\ &\times\left(\mathbf{H}^{\mathrm{T}}\mathbf{\Phi} + \frac{2\pi}{\lambda_L}\hat{k}^n_{\mathrm{mode}}\bar{\mathbf{D}} + \frac{4\pi}{\lambda_{\mathrm{Ku}}}\hat{k}_2|\mathbf{D}_{\mathrm{GB}}|\mathbf{G}^{\mathrm{T}}\right).\end{aligned} \tag{31}$$

Finally, the 3-D deformation fields are represented as follows:

$$\hat{\mathbf{D}}_{\mathrm{all2}}(x, y) = \mathrm{kriging}\left(\hat{\mathbf{D}}_{\mathrm{cls2}}\left(\bar{G}_b\right) \cup \hat{\mathbf{D}}_{ls}\left(\bar{G}_f\right)\right) \tag{32}$$

where (x, y) represents any point in the scene.

In addition, it should be noted that the resolutions of GB-SAR (range resolution 0.375 m and azimuth resolution 0.233°) are different from that of the GNSS system (range resolution 5 m and azimuth resolution 10 m). For a PS in a GNSS-based InBSAR system, in its resolution cell, there are many PSs observed by GB-SAR. Therefore, the average value of the deformation monitoring results of these PSs is used as the GB-SAR monitoring result corresponding to one PS in GNSS-based InBSAR.

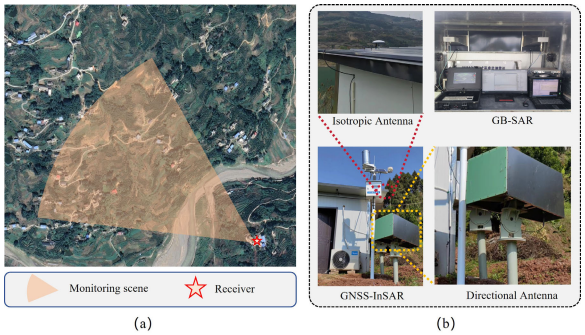


Fig. 6. Photographs of the scene and equipment in the experiment. (a) Optical image of the scene and (b) radar station and the equipment of GB-SAR and GNSS-based InBSAR including the directional antenna and isotropic antenna. The signal processor is placed in the radar station.

IV. Raw Data Processing

In this section, an experiment is conducted on a slope in Chongqing, China, as shown in Fig. 6(a). The scene contains a river, some houses, and roads. The slope and aspect are 22.5° and 142°. The area of the monitored scene is approximately 28.8×10^4 m^2. The GNSS-based InBSAR receiver and GB-SAR radar are deployed across the river as shown in Fig. 6(b). The longitude and latitude coordinates of the receiver are (109.4783E, 30.8869N). The aperture synthesis is provided by the movement of satellites. In order to achieve a certain azimuth resolution performance, the accumulation time is 600 s. In the GNSS-based InBSAR system, two antennas are used. The signal processor is placed in the radar station. Using the signal processor, the SAR image can be generated, and then is transmitted to the data center through the wireless network. Then, the PSs selection, phase error removal, multiangle SAR images association, and 3-D deformation retrieval process can be carried out. As for GB-SAR used in the experiment, the specific parameters of the equipment are shown in Table II. The experiment started on November 15, 2022, and lasted for 22 days using seven Beidou satellites in IGSO as transmitters in the GNSS-based InBSAR system. The specific experimental parameters, as shown in Table III. During the experiment, the GB-SAR radar is used to observe the scene simultaneously.

A. 3-D Deformation Fields Based on GNSS-Based InBSAR System

First, only the GNSS-based InBSAR is used to get the high-accuracy 3-D deformation monitoring results. The PS-InSAR technique is adopted here. After compensating the topographic phase and other error phases of PSs, deformation monitoring results of 21 days observed by each satellite, i.e., from different directions, can be obtained. The selection results of PSs by BDS2 IGSO2 and BDS3 IGSO1 satellites are shown in Fig. 7. The red points represent the PSs. It is clear that the selection

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TABLE II
PARAMETERS OF THE GB-SAR SYSTEM

Parameters	Value	Parameters	Value
Carrier frequency	16.2GHz	Band	Ku
Azimuth resolution	0.233°	Range resolution	0.375m
Synthetic aperture length	1.8m		

TABLE III
PARAMETERS OF THE GNSS-BASED INBSAR SYSTEM

Parameters	Value
Transmitter	BDS2 IGSO1,2,3,5,6 BDS3 IGSO1,3
Repeat-pass time	1 day
Bandwidth	10.23MHz
Equivalent PRT	1ms
Aperture time	600s
Carrier frequency	1268.52MHz (B3I)
Sampling frequency	100kHz
Horn antenna gain	15dB
Horn antenna polarization method	Horizontal polarization
Horn antenna main lobe width	42°
Isotropic antenna gain	2dB

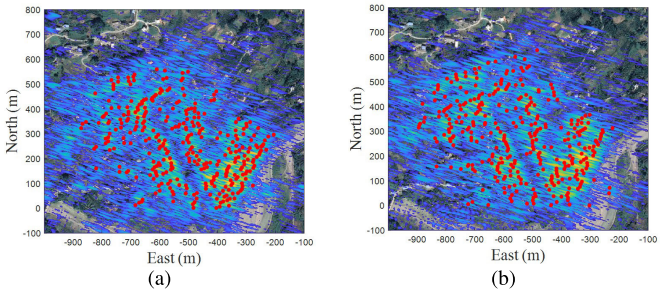


Fig. 7. PSs selection results. (a) Result of BDS2 IGSO2 satellite and (b) result of BDS3 IGSO1 satellite. The red points represent the PSs. It is clear that the selection results of PSs by different satellites are diverse. It is due to different satellite positions, that is, different monitoring angles of scene and bistatic configurations, making the RCS of target changes obviously under different satellite monitoring.

results of PSs from different satellites are different, especially in the northwest of the scene. It is due to different monitoring angles of the scene, i.e., different bistatic configurations, make the RCS of the target change obviously under different satellite monitoring.

To achieve 3-D deformation monitoring, the method proposed in [15] is used to associate the monitoring results from different satellites. The association result is shown in Fig. 8(a), where the PSs associated with 3–7 satellites are represented by red, green, blue, pink, and black, respectively. According to (9), the initial 3-D monitoring results can be obtained. Table I shows the corresponding monitoring accuracy and the number of PSs under different associated satellites. It can be seen that the 3-D deformation monitoring accuracy of GNSS-based InBSAR is closely related to the number of associated satellites. When the number of associated satellites is more than 6, the 3-D deformation monitoring accuracy remains within 10 mm. On the contrary, the monitoring accuracy deteriorates rapidly. Therefore, PSs with a low number of observing satellites need to be filtered.

Next, the 18th interference pair is taken as an example to obtain the deformation fields. The PSs are classified according

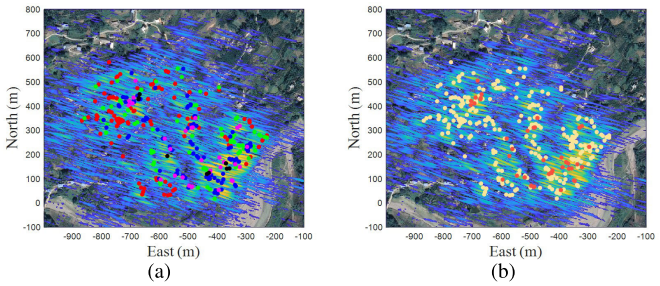


Fig. 8. Association results of SAR images. (a) Association results of multiple satellites, where the PSs associated with 3–7 satellites are represented as red, green, blue, pink, and black, respectively. PSs associated with six and seven satellites are defined as high-associated points $\bar{G}_f$, and the remaining PSs are defined as low-associated points $\bar{G}_b$. (b) Distribution of them. The 53 high-associated and 248 low-associated points are shown in red and white.

to the number of associated satellites. PSs associated with six and seven satellites are defined as high-associated points $\bar{G}_f$, and the remaining PSs are defined as low-associated points $\bar{G}_b$. Their distribution is shown in Fig. 8(b), in which 53 high-associated and 248 low-associated points are shown in red and white, respectively. The original 3-D deformation fields of the entire scene and their accuracy in east, north, and up directions are shown in Fig. 9 and Table I. The deformation monitoring accuracy of the scene is poor, and the deformation fields have large jitters. For example, the coordinate of the black point in Fig. 9 is $(-456, 447$ m), which is observed by three satellites, but two of them have similar elevation and azimuth angles. Therefore, the observation matrix $\mathbf{H}$ appears col-linearity and the 3-D deformation $\widehat{\mathbf{D}}_{\text{ls}}$ of this point is ill-posed and has a large deviation.

Subsequently, high-associated PSs are used to filter the scene 3-D monitoring results based on the spatial and temporal continuity of deformation. Taking the 3-D deformation results of high-associated PSs as input and using the Kriging interpolation method, the 3-D deformation of the entire scene after interpolation in east, north, and up directions are obtained, as shown in Fig. 10. The red points represent high-associated PSs. It is obvious that their deformation does not have large jitters due to a large number of satellites associated with these points, and the deformation observation accuracy is high which is 6.12, 4.24, and 5.93 mm, respectively.

After that, the deformation monitoring results of low-associated PSs are filtered. Taking the point at $(-500.6$, 303.4 m) as an instance, marked as point $\mathcal{A}$, which is a five-satellite association point, the initial 3-D deformation monitoring result $\widehat{\mathbf{D}}_{\text{ls}}$ is (21.33, 17.23, and 19.36 mm). To filter it, the expected deformation $\tilde{\mathbf{D}}_{\text{cls}}$ of this point is calculated by (19) based on the temporal continuity of deformation first. It is considered that the effective monitoring range r_{cls} is within 50 m near each PS. The expected deformation $\tilde{\mathbf{D}}_{\text{cls}}$ is (7.85 mm, 6.28 mm, 6.94 mm).

Afterward, the regularization parameter k_1 is selected through numerical simulation. Specifically, taking the expected deformation $\tilde{\mathbf{D}}^n_{\text{cls}}(q)$ as input, the interferometric phase of each satellite at point $\mathcal{A}$ is simulated by (20). Here, we take the noise standard deviation 0.186 rad as an example. The simulation range of k_1 is $[0 \sim 0.045]$. Subsequently, the estimation error corresponding to each k_1 is calculated, which is

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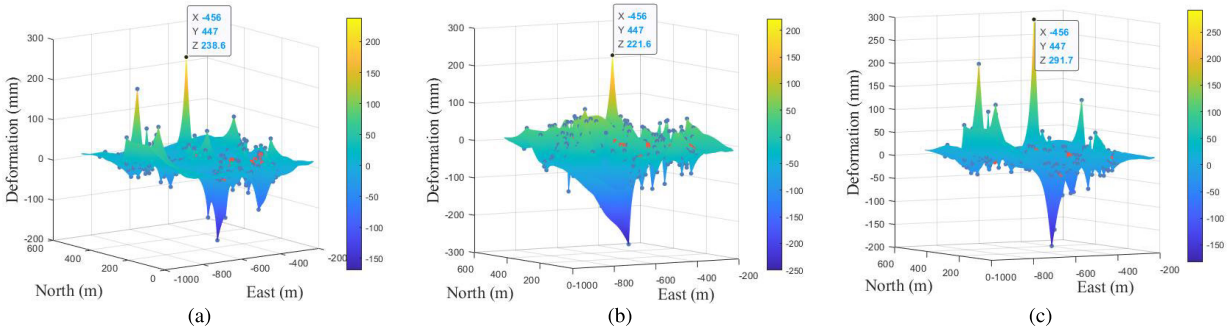


Fig. 9. Initial deformation fields of the entire scene. (a)–(c) Results in the east, north, and up directions, respectively. The black point is observed by three satellites. But two of the satellites are in similar elevation and azimuth angles. Therefore, the observation matrix appears collinearity. Therefore, the 3-D deformation at this point is ill-posed and has a large deviation.

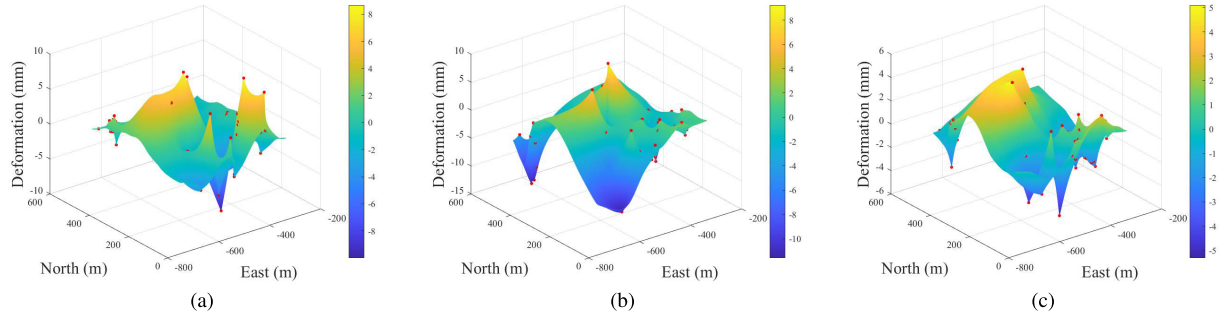


Fig. 10. Deformation of the entire scene interpolated by high-associated PSs of the 18th interference pair. (a)–(c) East, north, and up directions, respectively, and the red points represent the high-associated PSs.

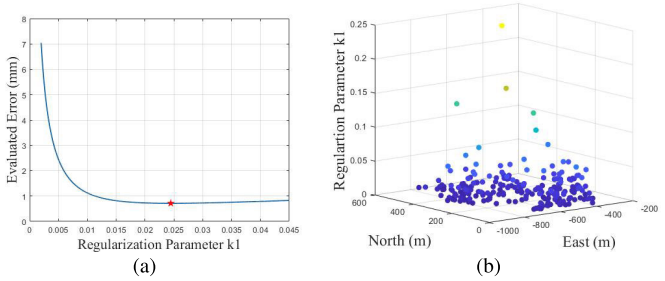


Fig. 11. (a) Relationship between the regularization parameter k_1 and evaluated error of the PS. (b) Regularization parameter $\hat{k}_1$ of the whole scene.

shown in Fig. 11(a). It can be seen that the corresponding estimation error is smallest when $\hat{k}_1 = 0.024$. Then, we conduct 100 Monte Carlo experiments by regenerating the noise randomly and solving for the corresponding $\hat{k}_1$. Subsequently, we choose the mode of them $\hat{k}_{1\text{mode}}$ as the regularization parameter of point $\mathcal{A}$ which is 0.024. Finally, the deformation monitoring results of point $\mathcal{A}$ after filtered $\widehat{\mathbf{D}}_{\text{cls}}$ are (8.65, 6.49, and 7.21 mm) which are calculated by (24). Compared with the initial deformation monitoring results, the filtered results have been greatly improved.

Using the same method, the regularization parameter estimation is performed on low-associated points in the entire scene, and the results are shown in Fig. 11(b). It is evident that the maximum variation range of $\hat{k}_{1\text{mode}}$ is close to 0.25, which is corresponding to the black point in Fig. 9. The initial deformation of several other PSs with large regularization parameter is all above 40 mm, so their k_1 are large. The deformation monitoring output error under different regularization parameters varies greatly. Therefore, it is necessary to optimize the regularization parameter.

After obtaining k_1 of the whole scene, the deformation fields can be obtained by (15). The results are shown in Fig. 12. It is obvious that after filtered based on deformation spatial continuity, the deformation jitters have been significantly reduced. In addition, the maximum and minimum values of scene deformation are reduced. The deformation of the black point in Fig. 9 has been filtered, which shows that high-associated points play an important role in the deformation measurement of low-associated points.

According to the number of associated satellites, the deformation of PSs before and after filtered is shown in Fig. 13. The left and right columns are the deformation before and after filtered correspondingly. From top to bottom are the results from the three-satellite to five-satellite association. It is evident that after filtered, the deformation measurement results of low-associated points have been significantly improved and the jitters are mitigated. The deformation monitoring accuracy of different associated satellite PSs and the numerical value of the accuracy improvement are shown in Table IV. The accuracy has also been corrected to less than 10.5 mm. This shows that the proposed algorithm can effectively improve the deformation monitoring accuracy of PSs with the lower number of associated satellites. In addition, it should be noted that the filtered deformation monitoring accuracy is not directly related to the number of observing satellites. Because it is affected by multiple factors, such as the position of PSs, historical deformation, and noise and is no longer solely dependent on the number of observing satellites.

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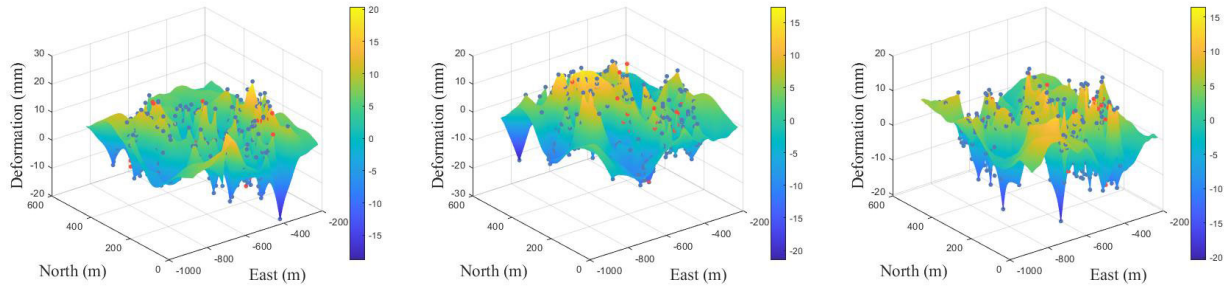


Fig. 12. Deformation fields of the whole scene after high-associated PSs filtered. (a)–(c) East, north, and up directions, respectively. The red and blue points represent the low-associated and high-associated PSs. The deformation jitters have been significantly reduced.

TABLE IV

THREE-DIMENSIONAL DEFORMATION MONITORING ACCURACY AND IMPROVEMENT VALUE OF 18TH INTERFERENCE PAIR AFTER HIGH-ASSOCIATED PSS FILTERED

Number of observing satellites	Accuracy (mm)			Number of PSs	Accuracy improvement (mm)		
	East	North	Up		East	North	Up
3	9.32	8.51	10.40	98	74.35	73.11	91.91
4	9.13	7.83	9.72	82	36.4	15.73	29.9
5	7.37	7.01	8.23	68	14.76	5.80	5.99
6	6.83	5.72	6.17	40	-	-	-
7	5.41	3.87	5.81	13	-	-	-
All of PSs	8.62	7.79	8.87	301	40.1	27.24	34.76

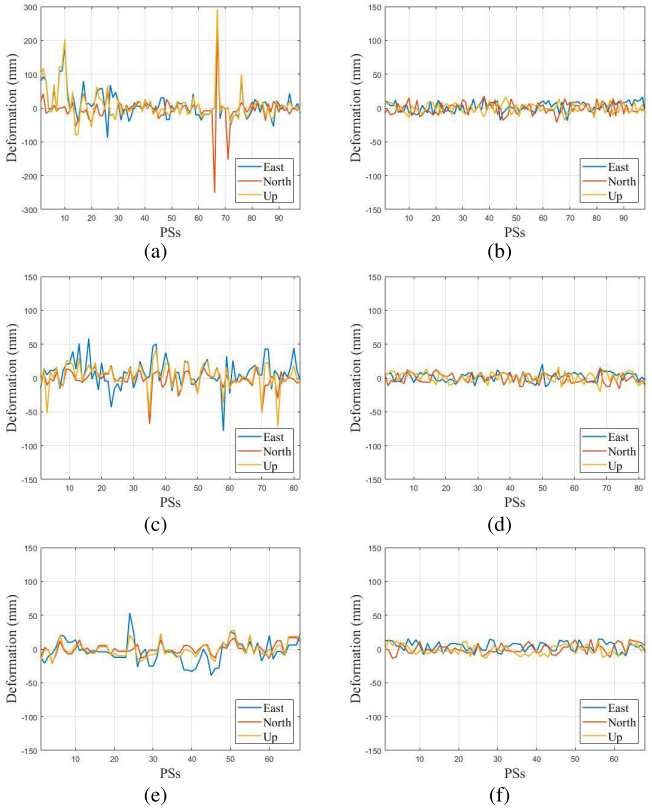


Fig. 13. Deformation of PSs before and after filtered. (a), (c), and (e) Deformation before filtered from three-satellite to five-satellite association, respectively. (b), (d), and (f) Deformation after filtered from three-satellite to five-satellite association, respectively.

For all 21 sets of interference pairs, the same processing method is used to obtain the final scene deformation fields under long-term monitoring. The final 3-D deformation measurement accuracy is shown in Table V, and the 3-D deformation monitoring accuracy is below 9 mm. Compared with the deformation accuracy before filtered, the deformation accuracy after filtered has been significantly improved.

TABLE V

22-DAY FULL-SCENE DEFORMATION MONITORING ACCURACY AFTER HIGH-ASSOCIATED PSS FILTERED

Accuracy (mm)	East	North	Up
Before corrected	50.21	38.03	45.17
After corrected	8.65	7.88	8.89
Improvement	41.56	30.15	36.28

In addition, it should be noted that the deformation monitoring accuracy is calculated by (10). Considering the observation period is relatively short, the nominal deformation is 0 upon assuming that no deformation occurred at the scene during the observation period.

So far, the 3-D high-accuracy deformation fields based on the GNSS-based InBSAR system are obtained. The initially obtained 3-D deformation fields have large jitters, and the deformation monitoring accuracy is poor. In addition, the deformation monitoring results of some PSs in the scene are ill posed. The algorithm proposed in this article can be considered as a filter specifically designed for GNSS-based InBSAR with multiple satellite observations. PSs with a higher number of observed satellites are given higher weights, while PSs with a lower number of observed satellites are assigned to lower weights. In this way, high-accuracy 3-D deformation field construction of the scene can be achieved without losing any observation information. The raw data verify the feasibility of the proposed algorithm in the GNSS-based InBSAR system and proves that the GNSS-based InBSAR system itself has millimeter-level accuracy deformation monitoring capability for actual scenes.

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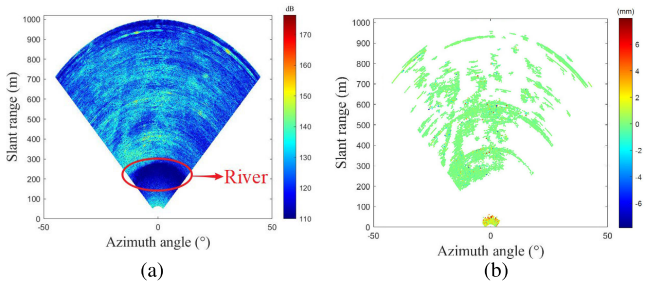


Fig. 14. (a) SAR image monitored by GB-SAR and the place with the lowest SNR is the river in the scene. (b) Deformation field of 18th day monitored by GB-SAR.

TABLE VI
ONE-DIMENSIONAL DEFORMATION RESULTS MONITORED BY GB-SAR AND GNSS-BASED INBSAR

	GB-SAR	GNSS-based InBSAR
Mean of deformation (mm)	0.22	0.86
RMSE	0.53	1.23

B. Verification of GNSS-Based InBSAR With GB-SAR 1-D Results

GB-SAR can obtain high-accuracy 1-D deformation monitoring results, which are used to verify the effectiveness of the GNSS-based InBSAR system. During the experiment, the GB-SAR radar is used to observe the same scene, and the SAR imaging result is shown in Fig. 14(a) at the polar coordinate. The place with the lowest SNR in Fig. 14(a) is the river in the scene. Then, the amplitude dispersion index and deformation monitoring algorithm [43], [44] are used to select PSs and obtain 1-D deformation fields in the GB-SAR system. A total of 53 078 PSs are selected, and the deformation field of the 18th day is obtained which are shown in Fig. 14(b). Then, the 3-D deformation results in GNSS-InBSAR are projected on the LOS of GBSAR according to (26). The statistical results of them are shown in Table VI. The mean values of the deformation monitoring results of the two systems are between 0.2 and 0.9, which are both very small. The RMSE value is within 0.5–1.5, and the RMSE of the GB-SAR is smaller than that of the GNSS-based InBSAR. The numerical statistical results suggest that both GB-SAR and GNSS-based InBSAR measure no deformation and landslides in the scene.

The deviation histogram distribution between GNSS-based InBSAR and GB-SAR 1-D deformation monitoring results is shown in Fig. 15. The deviation is distributed from -7 to 5 mm. The RMSE of the deviation is 1.37. It shows that the deviation between the GNSS-based InBSAR and GB-SAR monitoring results is small, which further proves the effectiveness and reliability of GNSS-based InBSAR deformation monitoring.

C. 3-D Deformation Fields Enhanced by GB-SAR

In Section IV-A, only GNSS-based InBSAR is used, and the deformation monitoring accuracy is below 9 mm. On this basis, the 3-D deformation monitoring accuracy of GNSS-based InBSAR is further improved by introducing the 1-D high-accuracy deformation observation of GB-SAR as the constraint.

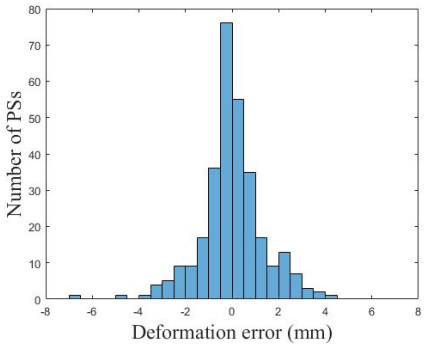


Fig. 15. Deviation histogram distribution between GNSS-based InBSAR and GB-SAR 1-D deformation monitoring results.

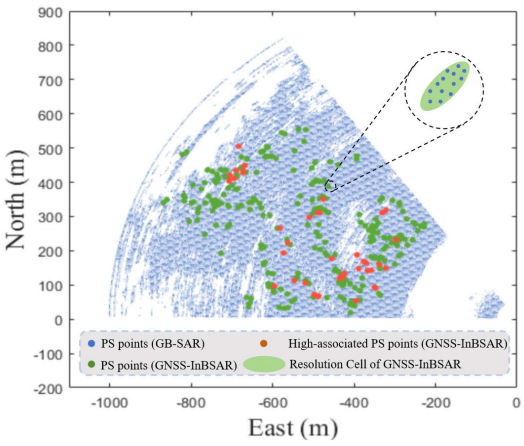


Fig. 16. PS selection results of GB-SAR and GNSS-based InBSAR after SAR images registration. The blue points represent the PS selection results of GB-SAR, the green points represent the PSs of the GNSS-based InBSAR system, and the red points represent the high-associated PSs in the GNSS-based InBSAR system.

As mentioned above, GB-SAR is used to observe the same scene, and the amplitude dispersion index method is used to select PSs in the GB-SAR system. The selection results of PSs in GB-SAR and GNSS-based InBSAR after SAR images registration at the east–north–up coordinate are shown in Fig. 16. The blue points represent the PSs in the GB-SAR system, the green points represent the PSs in the GNSS-based InBSAR system, and the red points represent the high-associated PSs in the GNSS-based InBSAR system. As mentioned in Section III-C, the range and azimuth resolutions are different between GNSS-based InBSAR and GB-SAR, so there is a difference in the density of PSs between the two systems.

Next, the 18th interference pair is taken as an example to demonstrate the proposed algorithm. Taking the point $\mathcal{A}$ as an example again, the deformation measurement results $\widehat{\mathbf{D}}_{\text{ls}}$ and $\widehat{\mathbf{D}}_{\text{cls}}$ of this point are (21.33, 17.23, and 19.36 mm) and (8.65, 6.49, and 7.21 mm). In its resolution cell, the deformation measurement result of GB-SAR is 0.4 mm. First, traverse the regularization parameter k_2 from 0 to 0.4 and solve the regression problem. Then, $\|\mathbf{\Phi} - \mathbf{H} \cdot \hat{\mathbf{D}}\|$ and $\|\mathbf{G} \cdot \hat{\mathbf{D}} - |\mathbf{D}_{\text{GB}}|\|$ versus each k_2 are calculated. Furthermore, the L-curve can be fit as shown in the blue line in Fig. 17(a).

The point with the largest curvature on the L-curve is the elbow point, which is shown as the red dot in Fig. 17(a). The corresponding regularization parameter is the optimal

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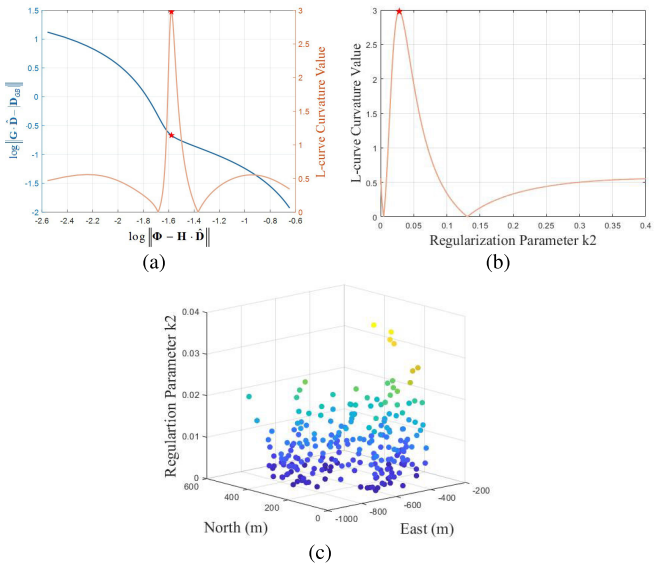


Fig. 17. (a) L-curve of PS in blue line and the curvature corresponding the L-curve in orange, (b) curvature of the L-curve corresponding to regularization parameter k_2, the corresponding $\hat{k}_2$ at the maximum curvature is 0.0282 as shown in the red dot in the figure, and (c) regularization parameter $\hat{k}_2$ of whole scene.

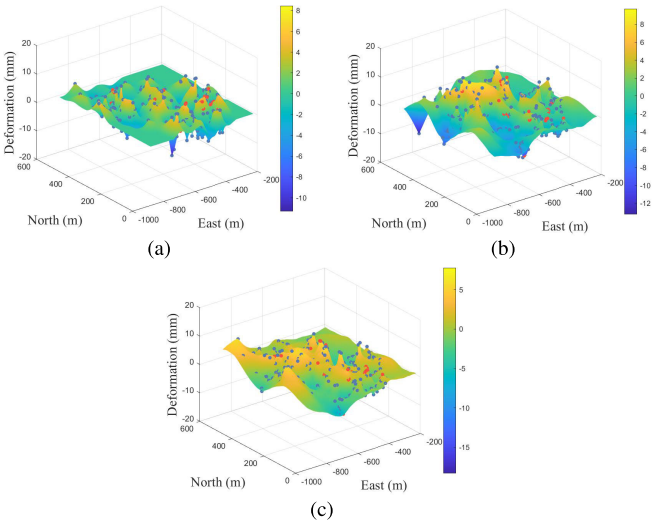


Fig. 18. Deformation fields of the whole scene after high-associated PSs filtered and GB-SAR deformation monitoring results constrained. (a)–(c) East, north, and up directions, respectively. The jitters of the deformation fields are further reduced.

k_2 for point $\mathcal{A}$ which is 0.0282. The deformation measurement result $\widehat{\mathbf{D}}_{\text{cls2}}$ is (4.8 mm, 4.5 mm, 6.47 mm). Compared with the deformation measurement result filtered by high-associated PSs $\widehat{\mathbf{D}}_{\text{cls}}$, the deformation result has been further improved.

Using the same method, $\hat{k}_2$ of the whole scene can be obtained as shown in Fig. 17(c). Then, the deformation fields after high-accuracy PSs filtered and GB-SAR constrained can be obtained by (32), as shown in Fig. 18. The jitters of the deformation fields are further reduced, and the deformation accuracy is further improved, as shown in Table VII. According to the number of associated satellites, the deformation before and after corrected is shown in Fig. 19. The left and right columns are the deformation before and after corrected correspondingly, and from top to bottom are the results

TABLE VII

THREE-DIMENSIONAL DEFORMATION MONITORING ACCURACY OF BEIDOU SATELLITES IN IGSO AFTER HIGH-ASSOCIATED PSS FILTERED AND GB-SAR CONSTRAINT

Number of observing satellites	Accuracy (mm) East	North	Up	Number of PSs
3	6.59	5.5	7.6	98
4	6.5	4.92	7.5	82
5	5.13	5.09	6.12	68
6	6.83	5.72	6.17	40
7	5.41	3.87	5.81	13
All of PSs	5.89	4.78	5.94	301

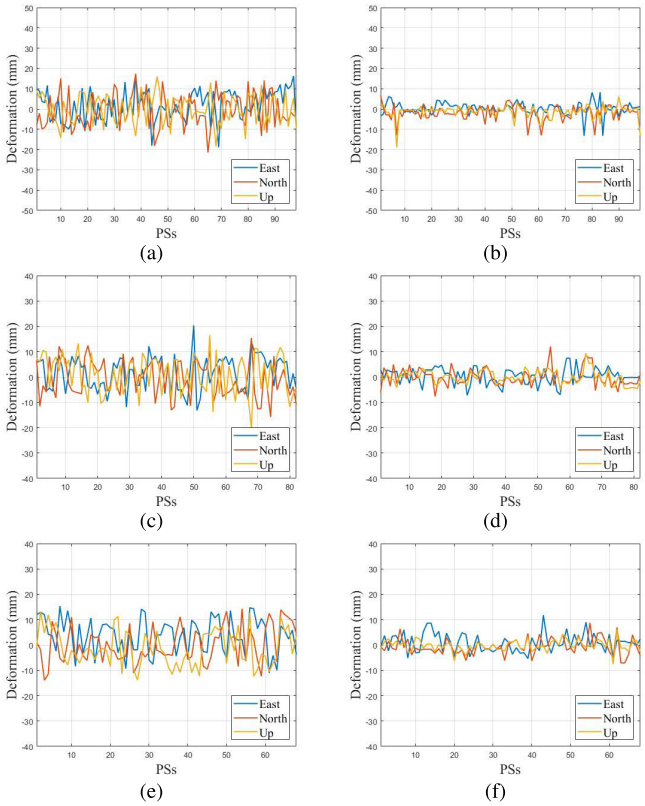


Fig. 19. Deformation of PSs before and after GB-SAR constrained. (a), (c), and (e) Deformation before processing from three-satellite to five-satellite association, respectively. (b), (d), and (f) Deformation after processing from three-satellite to five-satellite association, respectively. After being constrained, the deformation measurement of low-associated points has been significantly improved.

TABLE VIII

22-DAY FULL-SCENE DEFORMATION MONITORING ACCURACY AFTER HIGH-ASSOCIATED PSS FILTERED AND GB-SAR CONSTRAINED

Accuracy (mm)	East	North	Up
Before corrected	50.21	38.03	45.17
After first corrected	8.65	7.88	8.89
After second corrected	5.61	4.66	5.48

from the three-satellite to five-satellite association. It can be seen that after constrained, the deformation measurement of low-associated points has been significantly improved.

For the 21-day interference pairs, the same operation is performed to obtain the long-term deformation fields. The final 3-D deformation measurement accuracy is shown in Table VIII. Compared with the deformation fields filtered

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by the temporal and spatial continuity of deformation in Section IV-A, the accuracy of the deformation fields constrained by GB-SAR has been further improved from (8.65, 7.88, and 8.89 mm) to (5.61, 4.66, and 5.48 mm). It shows that GB-SAR plays an effective role in improving the deformation monitoring accuracy of GNSS-based InBSAR systems.

For GNSS-based InBSAR, 3-D deformation monitoring results can be obtained by associating different monitoring results from different satellites. However, the scene monitoring results are limited by the uneven associated 1-D monitoring results. Like the spatial filter adopted in the traditional space-borne InSAR results [45], the algorithm proposed in this article can also be considered as a filter. Different PSs with different multistatic configurations have different weights. Through the proposed algorithm, the advantages of multisatellite and multiangle observations in the GNSS-based InBSAR system can be fully utilized without losing any single-angle monitoring results. The raw data verify the feasibility of the proposed algorithm in GNSS-based InBSAR. On this basis, high-accuracy 1-D deformation results of GB-SAR are introduced to further improve the accuracy of 3-D deformation fields of GNSS-based InBSAR.

V. Conclusion

In this article, a 3-D deformation field construction algorithm based on spatial and temporal continuity is proposed for the GNSS-based InBSAR system. The first experimental 3-D deformation result of intersystem fusion is obtained. First, based on the spatial continuity, the PSs which can be observed by more satellites are used to filter the initial scene deformation monitoring results. In this process, the optimal regularization parameter is refined based on the temporal continuity of deformation. The 1-D deformation monitoring results of GB-SAR verify the effectiveness of GNSS-based InBSAR system deformation monitoring. On this basis, the 1-D monitoring results are introduced to further improve the monitoring accuracy of GNSS-based InBSAR. Finally, a 3-D deformation monitoring accuracy of within 6 mm is achieved. The experimental data verify the GNSS-based InBSAR system can obtain the 3-D deformation fields with high accuracy of the whole scene. Future efforts will focus on obtaining higher accuracy 3-D deformation fields using multisatellite resources.

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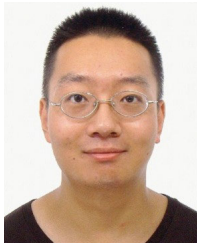
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